

Claire AI Full Data Model Map

****CLAIRE AI -- FULL DATA MODEL MAP****

All Objects: BO, PCO, PI, Signals, Scores, Clusters, Safety, Mode State

Medium-Wide Layout Unicode Symbols Only Export-Ready

1. Overview

This document defines the complete internal data model for Claire AI, including:

Breakthrough Object (BO)

Portfolio Candidate Object (PCO)

Portfolio Item (PI)

Signal Objects

Scoring Objects

Cluster Objects

Safety Objects

Mode State Objects

These schemas form the backbone of Claire's autonomous pipeline and are required for:

acquisition diligence

engineering implementation

safety governance

portfolio generation

mode switching

auditability

2. Breakthrough Object (BO)

Primary output of the Breakthrough Detection Engine

```
BreakthroughObject { id: UUID title: string description: string domain:
string created_at: ISO8601 mode_detected_in: deterministic | connected |
hybrid semantic_vectors: { concept_vector: float[] novelty_vector: float[]
feasibility_vector: float[] } signals: { tech_signals: TechSignal[]
market_signals: MarketSignal[] competitor_signals: CompetitorSignal[]
regulatory_signals: RegulatorySignal[] } scores: { novelty_score: float
feasibility_score: float market_alignment_score: float timing_score: float
competitive_pressure_score: float acquirer_fit_score: float risk_score:
float } safety: { redline_flags: string[] export_control_risk: boolean
```

```
dual_use_risk: boolean compliance_state: compliant | restricted | blocked
} status: draft | validated | blocked | promoted }
```

3. Portfolio Candidate Object (PCO)

Intermediate object created when a breakthrough passes thresholds

```
PortfolioCandidateObject { id: UUID breakthrough_id: UUID promoted_at:
ISO8601 rationale: { novelty: string feasibility: string market: string
timing: string strategic: string } scores: { composite_score: float
opportunity_score: float risk_score: float } safety: {
requires_governance_review: boolean redline_flags: string[] } status:
pending | approved | rejected | converted_to_portfolio_item }
```

4. Portfolio Item (PI)

Final asset created by the Portfolio Creator Engine

```
PortfolioItem { id: UUID title: string abstract: string created_at:
ISO8601 breakthrough_id: UUID pco_id: UUID technical: { architecture:
string components: string[] feasibility_model: string buildability_score:
float } market: { tam: float sam: float som: float competitive_landscape:
string adoption_timing: string market_rationale: string } strategic: {
acquirer_fit: string[] capability_gap_map: string strategic_timing: string
opportunity_window: string } risk: { technical_risk: float market_risk:
float regulatory_risk: float overall_risk_score: float } clustering: {
cluster_id: UUID theme_id: UUID adjacency_map: string[] } status: active |
archived | restricted }
```

5. Signal Objects

Used in Connected & Hybrid Modes

5.1 TechSignal

```
TechSignal { id: UUID source: string timestamp: ISO8601 content: string
relevance_score: float }
```

5.2 MarketSignal

```
MarketSignal { id: UUID source: string timestamp: ISO8601 content: string
trend_direction: up | down | neutral impact_score: float }
```

5.3 CompetitorSignal

```
CompetitorSignal { id: UUID competitor: string timestamp: ISO8601 action:
string impact_score: float }
```

5.4 RegulatorySignal

```
RegulatorySignal { id: UUID region: string timestamp: ISO8601 change_type:
update | proposal | enforcement severity: low | medium | high }
```

6. Scoring Objects

Used across breakthroughs, PCOs, and portfolio items

```
ScoreObject { novelty: float feasibility: float market_alignment: float
timing: float competitive_pressure: float acquirer_fit: float risk: float
composite: float }
```

7. Cluster Objects

Used by the Portfolio Engine

7.1 Cluster

```
Cluster { id: UUID name: string description: string portfolio_item_ids:
UUID[] created_at: ISO8601 }
```

7.2 Theme

```
Theme { id: UUID name: string description: string cluster_ids: UUID[] }
```

8. Safety Objects

Used by the Safety & Governance Layer

8.1 SafetyEvent

```
SafetyEvent { id: UUID timestamp: ISO8601 event_type: redline |
export_control | dual_use | governance_block description: string severity:
low | medium | high | critical mode: deterministic | connected | hybrid }
```

8.2 SafetyMatrixEntry

```
SafetyMatrixEntry { id: UUID category: string trigger_condition: string  
system_action: string requires_governance: boolean }
```

9. Mode State Object

Tracks current mode and engine states

```
ModeState { current_mode: deterministic | connected | hybrid last_switch:  
ISO8601 reason: string safety_state: compliant | restricted | blocked  
ingestion_active: boolean deterministic_engine_active: boolean  
connected_engine_active: boolean hybrid_fusion_layer_active: boolean }
```

10. Summary

This data model map:

defines every internal object

completes a major portion of the missing 20-30%

satisfies acquisition-grade diligence

enables engineering implementation

supports safety, governance, and auditability

proves Claire is a real, buildable system